

Topic: Topic modeling with language models (Langchain), **An LLM-Augmented Approach for Topic Modeling and Emotion Recognition in Vietnamese Online Education Platforms**

Statement:

Vietnamese NLP remains a low-resource area, especially for fine-grained emotion analysis. Existing models such as PhoBERT, VisoBERT, and CafeBERT achieve high accuracy for simple sentiment polarity, but their performance drops sharply when handling complex emotional categories. This affects real-world platforms where student-generated content includes nuanced, context-dependent expressions.

Large Language Models (LLMs) such as GPT-4 and GPT-5 demonstrate strong generalization and text understanding but are underexplored as augmentation or ensemble components in Vietnamese emotion classification. This research aims to fill that gap by integrating LLMs into a traditional transformer pipeline to enhance training data quality and final prediction reliability.

Why is it significant?

- LLMs have significantly developed in their ability to understand text and documents. Hence, they can be used to act as a topic modeling or emotion recognition model.
- Vietnamese NLP is considered a low-resource area, offering plenty of room for improvement.
- Available research either solely uses ML/deep learning models for both tasks or only employs prompt engineering with LLMs to achieve results.
- The need for a system that can perform state-of-the-art analysis and meaning extraction from student-generated content is urgently increasing due to the rapid growth of social media and technology.
- PhoBert and similar frameworks may perform excellently on simple emotion classifications (negative, neutral, positive). However, when moving to more detailed human emotions, performance drops significantly, from an accuracy of 0.9 to 0.5974 (with the 'other' label) and 0.6648 (without the 'other' label).

Research Objective:

- Build a high-performance fine-grained emotion recognition model using

transformer baselines (PhoBERT, VisoBERT, CafeBERT).

- Use LLMs (GPT-4/GPT-5) for controlled data augmentation to enrich low-resource emotion categories.
- Integrate LLM predictions into an ensemble framework to improve reliability and reduce misclassification of ambiguous emotions.
- Extract topics from texts using traditional and LLM-assisted methods to provide supporting context.
- Evaluate performance gains against strong baselines using standardized NLP metrics.

Research Questions:

- How much can LLM-based data augmentation improve fine-grained emotion recognition in Vietnamese educational text?
- Does an LLM-BERT ensemble outperform standalone transformer models in complex emotion classification?
- How does LLM-assisted topic extraction compare with traditional methods (LDA, BERTopic) in coherence and interpretability?
- What is the combined impact of augmentation + ensemble on model robustness across emotion categories?

Metrics:

- Macro F1, Micro F1
- Accuracy
- Coherence score (for topic modeling)

Methodologies:

- Baseline Models
 - Train and evaluate the following:
 - PhoBERT
 - VisoBERT
 - CafeBERT
 - Metrics:
 - Macro F1 (primary metric)
 - Micro F1

■ Accuracy

These provide a benchmark for all improvements.

- LLM-Based Data Augmentation (Primary Contribution)
 - Steps:
 1. Identify low-frequency emotion categories from the dataset.
 2. Use GPT-4/GPT-5 to generate synthetic training sentences:
 - Controlled prompts to maintain Vietnamese linguistic authenticity
 - Multiple emotional intensities
 - Avoiding label leakage and bias
 3. Validate synthetic data
 4. Retrain PhoBERT/VisoBERT/CafeBERT with original + synthetic data.
 5. Quantitatively evaluate improvements over baselines.
 - Expected outcome:
 - Higher macro-F1, especially for minority emotion classes.
- LLM-BERT Ensemble Model (Secondary Contribution)
 - Approach:
 1. Perform inference using:
 - Top 3 best BERT model
 2. Combine predictions using:
 - Confidence score of predictions
 3. Evaluate ensemble vs. individual models.
 - Expected outcome:
 - Reduced error on ambiguous emotional expressions and sarcasm/implicit emotions.
- Topic Extraction (Supporting Task)
 - Methods:
 1. Traditional baseline:
 - BERTopic (embedding-based)
 2. LLM-assisted topic generation:
 - LangChain map-reduce summarization
 - GPT-4/GPT-5 topic clustering

- Metrics:
 - Coherence score (Cv)
 - Human interpretability score

Goal: Provide contextual explanations that relate topic distributions to detected emotions.